
FINRCA: Anomaly-Propagation-Aware Lineage Graph for Automated Root Cause Analysis in Financial Risk Pipelines

Won Woo

Department of Computer Science
Hanyang University
thewoowon@hanyang.ac.kr

Abstract

Modern financial institutions are required under BCBS 239 to maintain complete, accurate, and traceable data lineage across their risk reporting pipelines. Yet as of 2025, fewer than 10% of globally systemically important banks fully comply, with data lineage cited as the most persistent challenge. When anomalies occur in multi-stage financial risk pipelines—spanning raw market data ingestion, ETL transformations, feature engineering, and regulatory reporting—identifying the root cause manually requires hours to days of investigation.

We propose FINRCA, a system that (1) automatically constructs a column-level **Financial Risk Lineage Graph (FRLG)** from a risk pipeline definition, and (2) runs an **Anomaly-Propagation-Aware Root Cause Analysis (APA-RCA)** algorithm on the graph to identify root causes of observed anomalies. APA-RCA extends Random Walk with Restart (RWR) with anomaly-score-weighted transitions, transform-type-based edge weights, and ancestor-constrained ranking. A multi-signal anomaly detector—combining causal excess isolation, peer deviation scoring, and short-long variance ratio analysis—provides the anomaly scores that drive the walk.

We evaluate FINRCA on the SYNFRP benchmark: a synthetic financial risk pipeline with controlled anomaly injection across seven anomaly types and three pipeline scales (45–100+ nodes, 630 total runs). APA-RCA v2 achieves **77.9% overall Top-5** across all anomaly types—outperforming Anomaly-Score Only (70.2%) by 7.7 pp ($\chi^2 = 30.4$, $p < 10^{-7}$) and achieving 100% on four of seven types. A companion *Schema-Constraint RCA* module resolves A4 aggregation errors via FRLG structural integrity checks, achieving 100% Top-1 with zero false positives on other anomaly types. The combined Hybrid achieves **91.9% Top-5** and **59.5% Top-1** ($\chi^2 = 111.7$, $p < 10^{-25}$ vs. Anomaly-Score Only). An expanded Real-Source Hybrid Benchmark (RSHB) covering 12 instruments (10 US equities, 2 Korean Exchange equities, FX, and Treasury rates) over 756 trading days spanning three market regimes achieves 82.4% Top-5 (+36.2 pp over Anomaly-Score Only), confirming robustness under fat-tail conditions and cross-market heterogeneity. Extended scalability confirms sub-20ms RCA query latency up to 1 009-node pipelines.

1 Introduction

1.1 The BCBS 239 Compliance Gap

In 2013, the Basel Committee on Banking Supervision published BCBS 239 Basel Committee on Banking Supervision [2013], mandating that banks ensure the accuracy, completeness, and timeliness of their risk data, and demonstrate end-to-end traceability from source systems to regulatory reports.

Over a decade later, the gap between regulation and practice remains striking. BIS’s 2025 implementation review Bank for International Settlements [2025] reports that among 31 G-SIBs, *only 2 institutions are in full compliance*, and no single principle has achieved universal compliance. The ECB’s 2024 RDARR guide European Central Bank [2024] explicitly names attribute-level data lineage as one of seven supervisory priority areas. PwC’s 2024 industry survey PricewaterhouseCoopers [2024] finds that structural data governance deficiencies persist across the majority of G-SIBs.

The technical difficulty is real: financial risk pipelines are multi-stage, with raw market data undergoing dozens of sequential transformations—cleaning, currency conversion, feature engineering, risk metric calculation—before reaching regulatory reports. Each stage introduces potential failure points. When a reported VaR figure is incorrect, finding which upstream transformation introduced the error requires traversing a complex dependency graph, a task that today remains largely manual.

FRTB 2025 raises the stakes further. The Fundamental Review of the Trading Book (FRTB), mandatory for G-SIBs from January 2025 Basel Committee on Banking Supervision [2019], substantially increases both the complexity and the auditability requirements of risk pipelines. Two FRTB provisions directly amplify the need for automated data lineage RCA. First, the *P&L Attribution Test* (PLAT) requires firms to demonstrate monthly that their Internal Model Approach (IMA) risk-theoretical P&L closely tracks the hypothetical P&L computed from front-office pricing. Failure causes automatic fallback to the Standardized Approach, which typically carries 40–60% higher capital requirements Basel Committee on Banking Supervision [2019]. A PLAT failure is, at its root, a data lineage discrepancy: some upstream transformation produced figures inconsistent with the front-office source. Identifying which node is the culprit is precisely the problem our system solves. Second, FRTB’s Sensitivity-Based Approach (SBA) mandates risk factor sensitivity calculations (delta, vega, curvature) at the individual instrument level, adding new computation layers to existing VaR pipelines—increasing node counts and the manual RCA burden proportionally. Our scalability experiments (Section 6) confirm that APA-RCA remains below 20 ms latency at 1 009-node pipeline scales, consistent with daily PLAT reporting windows.

1.2 Motivation

During employment at KB Kookmin Bank’s StarBanking Operations Team, the author directly observed the consequences of inadequate data lineage tooling. When discrepancies arose in reported risk metrics, operational teams spent hours to days tracing data flows across disparate systems, relying on tribal knowledge and ad hoc SQL queries. This experience motivates the central research question:

Can the root cause of a data quality anomaly in a financial risk reporting pipeline be identified automatically, rapidly, and accurately using a lineage graph?

1.3 Contributions

This paper makes six contributions:

C1—FRLG. We define the *Financial Risk Lineage Graph* schema for modeling column-level data lineage in financial risk pipelines. The schema captures five node types (Source, ETL, Feature, Risk, Report) and four edge types (DirectMap, Filter, Calculate, Aggregate), reflecting the structure of VaR/ES computation pipelines. To our knowledge, this is the first column-level lineage graph schema specifically designed for financial risk computation (as opposed to general-purpose metadata catalogs such as OpenLineage or Apache Atlas).

C2—Hop-Distance-Adaptive APA-RCA. We propose *Anomaly-Propagation-Aware Root Cause Analysis* (v2), which extends RWR with a novel hop-distance-adaptive weighting scheme: nodes close to the observed anomaly target receive higher graph-structure trust, while distant ancestors rely more on direct anomaly evidence. This is the first RCA method to adapt RWR restart probability based on hop distance in any domain; all prior graph-walk RCA methods (MicroRCA, CloudRanger, CORAL, MULAN, REASON) use fixed transition weights. A companion five-signal anomaly detector combines causal excess isolation, peer deviation scoring, short-long variance ratio, rolling z-score, and segment variance signals.

C3—Schema-Constraint RCA. We introduce a lightweight ($O(|V|)$) structural integrity check on the FRLG that detects aggregation schema violations (A4) with 100% Top-1 accuracy and zero false positives on non-A4 types—a complementary approach absent from all existing RCA literature.

C4—SYNFRP Benchmark. We construct the *Synthetic Financial Risk Pipeline* benchmark: a configurable pipeline with GBM/Vasicek/RW-based data generation and a controlled seven-type anomaly injection protocol with ground truth (630 runs). To our knowledge, this is the first public benchmark for evaluating RCA algorithms in the financial risk data lineage domain.

C5—Anomaly Transfer Function (ATF) Framework. We derive a formal model of anomaly propagation through financial pipeline transforms, yielding closed-form transfer coefficients ρ_τ per transform type (Section 3.4). The ATF provides principled values for APA-RCA’s hop attenuation ($\alpha^* \approx 0.72$, matching the hand-tuned $\alpha = 0.7$) and edge weights, explains why removing the hand-tuned β improves ablation performance, and derives a distribution-free temporal discriminant ϕ_{peer} that perfectly separates A3 code bugs from A5 spike propagation on both synthetic and real data (0% overlap).

C6—Real-Source Hybrid Benchmark (RSHB). We validate on an expanded real-market dataset comprising 12 instruments (US equities, Korean Exchange equities, FX, Treasury yields) over 756 trading days spanning three market regimes (Fed tightening, banking stress, AI rally). The RSHB is the first RCA evaluation benchmark built on actual financial market data from two distinct equity markets.

2 Related Work

Data lineage systems. Cheney et al. [2009] provide a foundational survey of data provenance in databases, distinguishing why-, where-, and how-provenance. Davidson and Freire [2008] survey provenance in scientific workflows. Industrial tools—Apache Atlas, OpenLineage, Atlan, Solidatus—provide lineage visualization and governance, but none offers RCA capability, and none has been evaluated with published RCA accuracy metrics. No academic work has addressed lineage-based RCA in the financial risk domain.

Root cause analysis. RCA has been extensively studied in cloud and microservices contexts. MonitorRank Kim et al. [2013] pioneered random walk on sensor dependency graphs at LinkedIn, establishing the core propagation-walk paradigm. CloudRanger Wang et al. [2018] applied second-order random walk on PC-algorithm causal graphs for cloud-native systems. MicroRCA Wu et al. [2020] introduced Personalized PageRank on attributed service topology graphs (89% precision), becoming the canonical graph-walk RCA reference. GROOT Wang et al. [2021] deployed GrootRank (a PageRank variant) on event causal graphs in production e-commerce at 78% Top-1. More recent methods include CIRCA Li et al. [2022] (causal Bayesian network with intervention recognition at KDD 2022), CORAL Wang et al. [2023b] and REASON Wang et al. [2023c] (both KDD 2023: incremental/hierarchical causal graph + RWR), MULAN Zheng et al. [2024] (multi-modal causal learning + RWR, WWW 2024), Nezha Yu et al. [2023] (multi-modal event graph, FSE 2023), and Chain-of-Event Yao et al. [2024] (weighted event causal graph from deployment history, FSE 2024 industry). RCACopilot Chen et al. [2024] uses LLMs for incident root-cause generation. BARO Pham et al. [2024] applies Bayesian change-point detection. AERCA Xiao et al. [2025] (ICLR 2025 Oral) uses Granger causal discovery to identify anomaly-introducing interventions in time series. RCAEval Pham et al. [2025] provides the most comprehensive benchmark. Wang and Qi Wang and Qi [2024] survey 100+ methods through 2024.

All of the above target *system failures* (service crashes, latency spikes, resource exhaustion) in microservice/cloud environments. None addresses data quality anomalies propagating through finan-

cial risk computation pipelines, and none uses data lineage graphs as the propagation substrate. Furthermore, while multiple methods use RWR or PageRank as the localization walk, none adapts the restart probability as a function of hop distance from the observed anomaly target—FINRCA’s primary algorithmic contribution.

Li et al. [2024] (GNN-based root cause localization for general data pipelines) and Assaad et al. [2023] (acyclic causal graph methods for time series anomalies) are the closest prior work in terms of problem framing, but neither targets financial regulatory pipelines nor provides a domain-specific anomaly taxonomy. Wang et al. [2023a] propose hierarchical GNNs for causal discovery. Our work differs in domain specificity (financial risk pipelines, BCBS 239), algorithm design (hop-distance-adaptive typed-edge RWR with multi-signal detector), and the provision of a reproducible benchmark with seven financial anomaly types and real-source validation across two equity markets (US + KRX).

Graph-based propagation. Tong et al. [2006] establish the theoretical foundations of RWR on graphs. APA-RCA extends RWR with three domain-specific modifications described in Section 4.3.

OpenLineage to FRLG schema mapping. A key practical concern is whether FINRCA can ingest lineage metadata produced by existing tools without manual re-engineering. Table 1 shows the field-level mapping between the OpenLineage RunEvent schema OpenLineage Community [2024] and the FRLG node/edge schema. Each OpenLineage Job (a transformation step) maps to a single FRLG node; each InputDataset/OutputDataset pair defines an edge. Column-level lineage facets (ColumnLineageDatasetFacet) map directly to FRLG `column_in/column_out` fields. The `transformType` field is inferred from the job’s `JobTypeJobFacet` (SQL `calculate`; aggregate SQL `aggregate`; reporting job `report`). This mapping confirms that FINRCA can be deployed on any OpenLineage-compliant pipeline without custom instrumentation.

Table 1: Field-level mapping from OpenLineage RunEvent schema to FRLG node and edge schema. * = inferred from job context; = populated by SynFRP ground truth in synthetic experiments.

OpenLineage Field	FRLG Target	Notes
<code>job.name</code>	<code>node_id</code>	Unique node identifier
<code>job.namespace</code>	<code>namespace</code>	Pipeline/team namespace
<code>JobTypeJobFacet.jobType</code>	<code>transform_type*</code>	SQL <code>calculate</code> , AGG <code>aggregate</code>
<code>run.runId</code>	<code>last_run_id</code>	For audit trail
<code>inputs[].name</code>	upstream edge source	(<code>src</code> , <code>dst</code>) edge
<code>outputs[].name</code>	downstream edge target	(<code>src</code> , <code>dst</code>) edge
<code>ColumnLineageDatasetFacet.fields</code>	<code>column_in/out</code>	Column-level lineage
<code>DataQualityMetricsInputFacet</code>	<code>anomaly_score</code>	If available; else SynFRP
<code>DataSourceDatasetFacet.sourceType</code>	<code>node_type</code>	SOURCE vs TRANSFORM

3 Problem Formulation

3.1 Financial Risk Lineage Graph (FRLG)

A *financial risk pipeline* $P = (N, D)$ consists of processing nodes N and directed data flow edges $D \subseteq N \times N$. Nodes are organized in five layers: **L1** Source, **L2** ETL, **L3** Feature, **L4** Risk, **L5** Report.

The *Financial Risk Lineage Graph* $\mathcal{G} = (V, E, \ell_V, \ell_E)$ is a labeled directed acyclic graph derived from P , where:

- $V = N, E = D$
- $\ell_V : V \rightarrow \{\text{Source, ETL, Feature, Risk, Report}\} \times [0, 1]$ assigns each node its type and anomaly score
- $\ell_E : E \rightarrow \{\text{DirectMap, Filter, Calculate, Aggregate}\}$ assigns each edge its transform type

Edge weight prior β . We assign structural weights based on transform type: Aggregate $\beta = 1.5$, Report $\beta = 1.3$, Calculate $\beta = 1.2$, DirectMap $\beta = 1.0$, Filter $\beta = 0.8$. These reflect the domain prior that aggregation and calculation transforms have higher causal significance.

3.2 Multi-Signal Anomaly Scoring

For each node $v \in V$ with output time series x_v , anomaly score $a(v) \in [0, 1]$ is computed by a five-signal detector.

Rolling z-score (S1). $s_z(v) = 0.6 \cdot \frac{|\{t: |z(t)| > \theta\}|}{T} + 0.4 \cdot \min\left(\frac{\max_t |z(t)|}{3\theta}, 1\right)$ where $z(t) = (x(t) - \mu_{\text{roll}}(t))/\sigma_{\text{roll}}(t)$ and $\theta = 2.0$.

Short-long variance ratio (S2). $s_{\text{sl}}(v) = \min\left(\frac{\sigma_5(v)/(\sigma_{30}(v)+\varepsilon)-1}{5}, 1\right)$ where σ_k is the median k -day rolling standard deviation computed on the warm series (excluding the first $\min(w, T/4)$ initialization steps).

Segment variance (S3). Split x_v into four equal quarters; $s_v(v) = 0.8 \cdot \min((\sigma_{\text{max}}/\sigma_{\text{min}} - 1)/10, 1)$.

Peer deviation with CV normalization (S4). Group nodes by peer key (layer, metric_type, instrument_family). Within each group, compute the Coefficient of Variation $\text{CV}(v) = \sigma(x_v)/(|\mu(x_v)| + \varepsilon)$ for scale-invariant comparison. For groups with ≥ 2 peers, $s_{\text{peer}}(v) = \min(|\text{CV}(v) - \text{CV}|/(\text{std CV} + \varepsilon)/5, 1)$.

Causal excess (S5). $\text{excess}(v) = a_{\text{raw}}(v) - 0.7 \cdot \max_{u \in \text{upstream}(v)} a_{\text{raw}}(u)$, where a_{raw} uses only S1–S3. Source nodes with no upstream receive $\text{excess} = a_{\text{raw}}$. $s_{\text{causal}}(v) = \max(0, \text{excess}(v)) + 0.2 \cdot \mathbb{1}[\text{excess}(v)/a_{\text{raw}}(v) > 0.8]$.

Layer-aware fusion.

$$\begin{aligned} a(v) &= \max(s_z, s_{\text{sl}} \cdot 0.5, s_{\text{peer}} \cdot 0.75) \quad \text{if } \text{layer}(v) \in \{1, 2\} \\ a(v) &= \max(s_z \cdot 0.55, s_{\text{causal}} \cdot 0.85, s_{\text{sl}} \cdot 0.80, s_{\text{peer_causal}} \cdot 0.90) \quad \text{if } \text{layer}(v) = 3 \\ a(v) &= \max(s_z \cdot 0.40, s_{\text{causal}} \cdot 0.60, s_{\text{peer}} \cdot 0.35) \quad \text{if } \text{layer}(v) \in \{4, 5\} \end{aligned}$$

where $s_{\text{peer_causal}} = s_{\text{peer}} \cdot \min(\text{excess}/0.3, 1)$.

3.3 RCA Problem

Input: $\mathcal{G}$, anomaly scores $A : V \rightarrow [0, 1]$, target node $t \in V$ (anomalous report node). **Output:** Ranked list $R = (r_1, r_2, \dots)$ with the true root cause r^* as high as possible. Performance is measured by the rank of r^* in R .

3.4 Anomaly Transfer Function (ATF) Framework

We now derive a principled model of how anomalies propagate through financial pipeline transforms. The *Anomaly Transfer Function* (ATF) for a transform f_τ with type τ measures the z-score transfer ratio:

$$\rho_\tau = \frac{\|f_\tau(x + \delta) - f_\tau(x)\|_z}{\|\delta\|_z} \quad (1)$$

where $\|s\|_z = \max_t |s_t - \mu_s|/\sigma_s$ is the max z-score of a series. ρ_τ quantifies what fraction of an input anomaly's z-score magnitude survives the transform.

Per-transform derivation.

- **Source / Direct-map ($\rho = 1.0$):** Identity or scalar multiplication; anomaly passes through unchanged.

- **Filter clip at c std** ($\rho = \min(1, c/z_{\text{in}})$): Inputs within the clip boundary pass through ($\rho = 1$); larger spikes are truncated. For $c = 4$ and a spike of $z = 8$: $\rho = 0.5$. This is the formal model of the A5 masking mechanism.
- **Calculate log-return** ($\rho \approx 1.0$): $f(x)_t = \log(x_t/x_{t-1})$; a spike at t_0 produces ATF non-zero at both t_0 and t_0+1 (the derivative doubles the temporal footprint).
- **Aggregate mean of k inputs** ($\rho = 1/\sqrt{k}$): For i.i.d. inputs with one anomalous: $\sigma_{\text{out}} = \sigma_{\text{in}}/\sqrt{k}$, so the z-score ratio is $1/\sqrt{k}$. For $k = 5$ (medium pipeline): $\rho \approx 0.45$.
- **Aggregate rolling std of window w** : A transient spike affects w of T time steps; $\rho_{\text{spike}} \approx 1/\sqrt{w}$. A persistent level shift (code bug) affects all T steps: $\rho \rightarrow 1$.
- **Report binary threshold** ($\rho \approx 0$): Discontinuous output carries near-zero z-score information, explaining why REPORT_LIMIT scores 0 in the RSHB (Section 6.7).

Connection to APA-RCA parameters. The ATF framework provides principled values for the edge weights β and hop attenuation α in the RWR transition matrix:

$$\beta_{\tau}^* = \rho_{\tau} \quad (\text{high-transfer edges more likely to carry anomaly}) \quad (2)$$

$$\alpha^* = \bar{\rho}^{-1} \approx 0.72 \quad (\text{geometric mean of per-hop } \rho) \quad (3)$$

The hand-tuned $\alpha = 0.7$ matches the ATF prediction to within 3%. In contrast, the hand-tuned $\beta_{\text{aggregate}} = 1.5$ is *opposite* to the ATF prediction of $\rho = 0.45$ —aggregation dilutes anomalies, so backward walk weight should be *low*, not high. This explains the ablation result that removing β improves performance (Section 6.3): uniform $\beta = 1.0$ is closer to theory than the misaligned hand-tuned values.

ATF-derived causal excess. The causal excess signal (S5) uses a global discount factor of 0.7. ATF derives an edge-specific discount:

$$\text{excess}^*(v) = a(v) - \rho_{\tau(v)} \cdot \max_{u \in \text{up}(v)} a(u) \quad (4)$$

For aggregate nodes ($\rho = 0.45$), a smaller discount means the node retains more credit as a potential origin—appropriate because aggregation strongly dilutes propagated anomalies, so any residual score is more likely to be locally generated.

Temporal fraction ϕ_{peer} : distribution-free A3/A5 discriminant. The ATF predicts distinct temporal signatures for code bugs versus data spikes: a wrong-window bug (A3) replaces $f \rightarrow f'$, affecting *all* T time steps; a spike (A5) propagates through a window of width w , affecting w/T steps. We define the *peer temporal fraction*:

$$\phi_{\text{peer}}(v) = \frac{1}{T} \left| \left\{ t : \frac{|x_v(t) - \bar{x}_{\text{peer}}(t)|}{\sigma_{\text{peer}}(t)} > 2 \right\} \right| \quad (5)$$

Empirically: $\phi_{\text{peer}}^{\text{A3}} \approx 0.7\text{--}0.9$ vs. $\phi_{\text{peer}}^{\text{A5}} \approx 0.05\text{--}0.37$, with zero overlap (Table 2). This replaces the fragile distribution-dependent $\text{raw} < 0.30$ gate with a principled, distribution-invariant signal. We gate the `peer_origin` detector with $\phi_{\text{peer}} > 0.5$: above this threshold, the anomaly is persistent (code bug); below, it is transient (spike propagation).

Table 2: ϕ_{peer} validation (30 trials each, medium pipeline). A3 (code bug) and A5 (spike propagation) are perfectly separated on both synthetic and real data.

Data	Type	Mean	Min	Max	Std
Synthetic	A3 (code bug)	0.794	0.452	0.940	0.125
Synthetic	A5 (spike)	0.195	0.012	0.413	0.095
Real	A3 (code bug)	0.727	0.548	0.864	0.089
Real	A5 (spike)	0.140	0.009	0.333	0.100

4 FINRCA System Design

FINRCA consists of four components: FRLG Builder, Multi-Signal Anomaly Detector, APA-RCA Engine, and Query Interface.

4.1 FRLG Builder

The FRLG Builder parses the pipeline definition and constructs $\mathcal{G}$ in three phases: (1) *node registration*—one vertex per pipeline node with attributes $\{\text{node_id}, \text{layer}, \text{transform_type}, \text{in_cols}, \text{out_cols}\}$; (2) *edge construction*—directed edges following data dependencies, with edge type assigned from the consuming node’s transform type and weight prior β ; (3) *DAG validation*—acyclicity check and column-reference validation.

The builder exposes four query types: forward impact query ($\text{desc}(v) = \text{nx.descendants}(\mathcal{G}, v)$), backward causality query ($\text{nx.ancestors}(\mathcal{G}, t)$), shortest causal path ($\text{nx.shortest_path}(\mathcal{G}, r, t)$), and full path enumeration for audit.

4.2 Target Node Selection

The system selects target t via priority ordering:

1. **L5 (Report):** $t = \arg \max_{r \in \text{Reports}} [a(r) + 0.1 \cdot |\{v \in \text{anc}(r) : a(v) \geq 0.1\}|]$
2. **L4 (Risk) fallback:** same formula with coefficient 0.05, applied if no report node exceeds the anomaly threshold.
3. **Global fallback:** most anomalous non-source node.

The L4 fallback is essential for A3 anomalies where feature errors do not consistently propagate above the L5 reporting threshold.

4.3 APA-RCA Algorithm

Input: $\mathcal{G}$, scores A , target t , restart prob γ , attenuation α .

Step 1—Transpose. Compute $\mathcal{G}^T$ (reverse all edges) to enable backward traversal from t toward root causes.

Step 2—Hop distances. BFS from t in $\mathcal{G}^T$: $h(v)$ = shortest path length from t to v in $\mathcal{G}^T$.

Step 3—Edge weight matrix. For each edge $(u \rightarrow v)$ in $\mathcal{G}^T$ (u downstream, v upstream):

$$w(u \rightarrow v) = a(v) \cdot \beta(\text{type}(u, v)) \cdot \alpha^{h(v)/h_{\max}} \quad (6)$$

Step 4—Transition matrix. Column-normalize W : $P_{ij} = W_{ij} / \sum_i W_{ij}$.

Step 5—Restart distribution.

$$e(v) = \begin{cases} 0.7 & v = t \\ 0.3 \cdot a(v) / Z & a(v) > 0.3, v \neq t \\ 0 & \text{otherwise} \end{cases} \quad (7)$$

where Z normalizes the secondary restart mass.

Step 6—RWR iteration. $r^{(k+1)} = (1 - \gamma)Pr^{(k)} + \gamma e$, until $\|r^{(k+1)} - r^{(k)}\|_1 < \varepsilon$ ($\varepsilon = 10^{-8}$).

Step 7—Final scoring. For each $v \neq t$, define hop-distance-adaptive graph/score weights and a score-inversed transform bonus:

$$w_{\text{graph}}(v) = 0.3 + 0.4 \cdot \left(1 - \frac{h(v)}{h_{\max}}\right), \quad w_{\text{score}}(v) = 1 - w_{\text{graph}}(v) \quad (8)$$

$$\beta_{\text{eff}}(v) = \max(\beta_{\text{base}}(\text{type}(v)) \cdot (1 - 0.5 \cdot a(v)), 0.1) \quad (9)$$

where $h(v)$ is the hop distance from t to v on $\mathcal{G}^T$. Nodes close to t (small h) receive higher graph weight—the RWR stationary mass is dense near the target, making the structural signal reliable. Conversely, at large hop distances the walk mass is diluted across many ancestors, so direct anomaly evidence outweighs the attenuated structural signal. The final composite score is:

$$\text{score}(v) = (w_{\text{graph}}(v) \cdot r(v) + w_{\text{score}}(v) \cdot a(v)) \cdot \text{anc}(v) \cdot \text{lb}(v) \cdot \beta_{\text{eff}}(v) \quad (10)$$

where $\text{anc}(v) = 1.0$ if $v \in \text{ancestors}(t)$ else 0.05 (ancestor constraint), and $\text{lb}(v) = 1 + 0.15(5 - \text{layer}(v))$ (upstream preference). The fixed-weight variant $0.4 \cdot r(v) + 0.6 \cdot a(v)$ (v1 baseline) is evaluated in the ablation study (Section 6).

Complexity. $O(|V| \cdot T)$ for anomaly detection (T = time series length), $O(K \cdot |E|)$ for RWR ($K < 50$ in all experiments). Total sub-2 ms for up to 100-node pipelines.

Relationship to standard RWR. APA-RCA extends standard RWR in three ways: (1) *transposed walk* on $\mathcal{G}^T$ (standard RWR operates on $\mathcal{G}$), (2) *anomaly-weighted transitions* scaling each edge by $a(v) \cdot \beta$, and (3) *composite ancestor-constrained scoring* that overrides the raw stationary distribution with domain structural knowledge.

5 SynFRP Benchmark

5.1 Design

SYNFRP is motivated by the absence of public financial pipeline datasets with known ground truth. It satisfies five requirements: realistic (GBM/Vasicek/RW models, five-layer pipeline structure), controlled (exact anomaly ground truth), reproducible (seeded RNG throughout), scalable (three size configurations), and representative (seven anomaly types covering the most common production failure modes).

5.2 Pipeline Topology

Table 3: SynFRP size configurations.

Configuration	Stocks	Rates	FX	Positions	Total Nodes
Small	3	2	1	3	≈ 45
Medium	5	3	2	5	≈ 63
Large	8	3	3	8	$\approx 100+$

The five-layer structure mirrors actual VaR/ES calculation workflows: **L1** sources (GBM stock prices, Vasicek rates, RW FX, slow-varying positions); **L2** ETL (forward-fill imputation, 4σ clipping, FX conversion, portfolio value join); **L3** features (log returns, 20-day and 60-day rolling volatility, sector aggregation, total portfolio value); **L4** risk metrics (VaR_{95} , VaR_{99} , ES_{95} , portfolio volatility, stress P&L, concentration risk); **L5** reports (regulatory VaR, internal dashboard, stress summary, limit breach flag).

5.3 Anomaly Types

Seven types model realistic financial pipeline failure modes:

A1—Source Corruption. Spike of magnitude $M = 8$ injected at a source price node for 1–5 days. Models vendor data feed errors. Propagation: price spike cascades through ETL, feature, and risk layers.

A2—ETL Logic Error. Scale factor M applied to FX conversion output, plus 10-day spike. Models misconfigured conversion rates. Peer deviation signal identifies the ETL node as an outlier among FX conversion peers.

A3—Feature Calculation Error. Volatility window reduced from 20 to $\max(2, \lfloor 20/M \rfloor) = 2$ days; output scaled by M . Models a window-parameter code bug. Detection relies on short-long variance ratio (S2) and peer deviation (S4).

A4—Aggregation Error. One instrument omitted from sector aggregation. Models an off-by-one instrument universe bug. Effect size $\approx 1/(N - 1)$ relative magnitude; typically insufficient to rank above background noise.

A5—Downstream Masking. Source spike just above the ETL 4σ clipping boundary (clip factor 3.8σ). The ETL clipping node absorbs 70–90% of the spike energy using a 60-day warmup window for boundary estimation, reducing source anomaly score to 0.35–0.50. Downstream feature nodes exhibit the highest apparent scores, creating a masking effect.

A6—Multi-Source Ambiguity. Two to three source nodes simultaneously receive anomalies with decaying magnitudes ($1.0, 0.6, 0.35\times$); their contributions aggregate at a risk node, which records the highest apparent score. Ground truth is the largest-magnitude source.

A7—Coincidental Anomaly. A legitimate source spike (root cause) is injected alongside market-driven noise at an unrelated pipeline branch, giving that noise node an anomaly score of $0.70\text{--}0.90$. The ancestor constraint eliminates the coincidental node ($0.05\times$ penalty).

Industry precedent for A5–A7. Each new scenario corresponds to documented production failure modes (Table 4).

Table 4: Industry precedent for new SynFRP anomaly types A5–A7.

Type	SynFRP Mechanism	Documented Industry Precedent
A5	ETL clipping absorbs spike	ECB RDARR European Central Bank [2024]: ETL normalization masking upstream data quality events
A6	Multi-source aggregation ambiguity	BCBS 239 Basel Committee on Banking Supervision [2013]: Principle 3 (accuracy) requires attribution of portfolio-level deviations to individual contributing positions
A7	Coincidental market-driven anomaly	PwC 2024 PricewaterhouseCoopers [2024]: 68% of false RCA incidents attributed to market volatility noise

Experimental design. $7 \text{ types} \times 3 \text{ sizes} \times 10 \text{ trials} = 210$ runs; each trial uses distinct seeds for data generation and injection point selection. Magnitude $M = 8.0$ fixed throughout.

6 Experiments

6.1 Setup

Implementation: Python 3.11, NetworkX 3.x, NumPy/Pandas. MacBook with Apple Silicon; all timing is wall-clock.

Hyperparameters: $\gamma = 0.15$, $\alpha = 0.7$, $\theta = 2.0$, $\varepsilon = 10^{-8}$, max iterations 1000.

Baselines: **BFS**—backward BFS from t on $\mathcal{G}^T$, ranked by hop distance (ties broken by anomaly score); **Vanilla RWR**—uniform transition weights, restart at t only, no ancestor constraint; **Anomaly-Score Only**—rank all non-target nodes by anomaly score descending, ignoring graph structure.

Metrics: Top- K accuracy ($K \in \{1, 3, 5\}$), MRR, False RC Rate@5, query latency (ms).

6.2 Main Results

Table 5 presents overall performance across all 630 runs (30 trials per anomaly type per pipeline size). APA-RCA v2 achieves **77.9%** Top-5 overall, outperforming Anomaly-Score Only (70.2%) by 7.7 pp ($\chi^2 = 30.4$, $p < 10^{-7}$, McNemar) and BFS (10.0%) by 67.9 pp. When augmented with Schema-Constraint RCA for A4 (Section 6.6), the Hybrid system achieves **91.9%** Top-5 and **59.5%** Top-1.

APA-RCA v2: adaptive weighting. The v2 improvements—adaptive graph/score weighting (Eq. 8) and score-inversed transform bonus (Eq. 9; fixed β removed from transition matrix in v2)—achieve **77.9% Top-5**, **44.4% Top-1**, and **MRR 0.605** on 630 runs versus v1’s 69.2% / 16.8% / 0.393. Augmented with Schema-Constraint RCA for A4 (Section 6.6), the Hybrid reaches **91.9% Top-5**, **59.5% Top-1**, **MRR 0.747**. The ablation study (Table 7) quantifies each v2 component’s contribution.

Table 5: Main results (630 runs across all 7 anomaly types and 3 pipeline sizes, 30 trials each). Best result in **bold**.

Method	Top-1	Top-3	Top-5	MRR	FRC@5	Latency (ms)
APA-RCA v2 + SC (Hybrid)	0.595	0.897	0.919	0.747	0.840	1.06
APA-RCA v2	0.444	0.754	0.779	0.605	0.849	0.70
APA-RCA v1	0.168	0.560	0.692	0.393	0.862	0.81
Anomaly-Score Only	0.249	0.632	0.702	0.446	0.865	0.01
BFS	0.000	0.057	0.100	0.069	0.980	0.66
Vanilla RWR	0.000	0.014	0.068	0.054	0.986	0.65
Structural-RWR	0.000	0.000	0.006	0.050	0.999	0.73
PC+RWR (CloudRanger-style)	0.000	0.000	0.006	0.050	0.999	7.23

Structural-RWR and PC+RWR. Two additional graph-walk baselines isolate the contribution of APA-RCA’s anomaly-weighted transition design. *Structural-RWR* uses uniform transition weights but biases the restart vector toward anomalous nodes—the approach taken by MicroRCA Wu et al. [2020] and related methods. *PC+RWR* replaces the domain-knowledge FRLG with a causal graph discovered by the PC algorithm ? from node time series (Fisher-Z independence test, $\alpha = 0.05$, depth 2), then applies the same anomaly-biased restart as Structural-RWR. Both achieve 0.6% Top-5—lower than even Vanilla RWR (6.7%)—because biasing the restart vector toward high-score nodes concentrates probability mass on downstream symptom nodes rather than upstream root causes. This confirms that the critical design choice in APA-RCA is *anomaly-weighting the transition matrix*, not the restart vector: directing the walk toward anomalous nodes at every step, rather than restarting from them, is what enables correct traversal through masking transformations (A5).

Anomaly-Score Only baseline analysis. Anomaly-Score Only achieves 70.2% Top-5. However, in scenarios where source signals are masked (A5, 10.0% for AO vs. 86.7% for v2), multiple sources are corrupted simultaneously (A6), or coincidental anomalies inflate off-path scores (A7), anomaly score alone is insufficient. APA-RCA v2’s advantage is confirmed by McNemar’s test on 630 paired runs: v2 vs. Anomaly-Score Only, $\chi^2 = 30.4$, $p < 10^{-7}$. The Hybrid vs. Anomaly-Score Only gap is even larger ($\chi^2 = 111.7$, $p < 10^{-25}$).

New scenarios (A5–A7) validate graph-guided RCA. Table 6 rows A5–A7 reveal the conditions under which graph structure provides essential disambiguation. In A5 (Downstream Masking), ETL clipping absorbs 70–90% of the source spike energy; APA-RCA v2 achieves 86.7% Top-5 on medium versus 10.0% for Anomaly-Score Only (+76.7 pp), because the ancestor-constrained walk correctly traces the residual signal through the clipping node back to the source. In A7 (Coincidental Anomaly), the ancestor constraint ($0.05 \times$ penalty for non-ancestors) eliminates market-noise nodes from the top candidates, yielding 100% versus 100% for Anomaly-Score Only (both achieve perfect accuracy, but v2’s ancestor constraint provides a cleaner ranking). These results directly demonstrate the value of lineage graph guidance in production-realistic anomaly scenarios.

Table 6 shows per-type breakdown on the medium pipeline (30 trials each). APA-RCA v2 achieves 100% on A1, A2, A6, and A7, with A5 at 86.7% and A3 at 90.0% via the ATF-derived ϕ_{peer} temporal discriminant. The Hybrid adds A4=100% via Schema-Constraint. Figure 1 visualizes these results.

Table 6: Top-5 accuracy by anomaly type (medium pipeline, 30 trials each). A5=Downstream Masking, A6=Multi-Source Ambiguity, A7=Coincidental Anomaly. SC=Schema-Constraint used for A4 in Hybrid row.

Method	A1	A2	A3	A4	A5	A6	A7
APA-RCA v2 + SC (Hybrid)	100%	100%	90.0%	100%	86.7%	100%	100%
APA-RCA v2	100%	100%	90.0%	0%	86.7%	100%	100%
APA-RCA v1	100%	93.3%	43.3%	0%	53.3%	90%	100%
Anomaly-Score Only	100%	100%	83.3%	6.7%	10.0%	100%	100%
BFS	0%	0%	83.3%	0%	0%	0%	0%
Vanilla RWR	0%	0%	40.0%	0%	0%	0%	0%

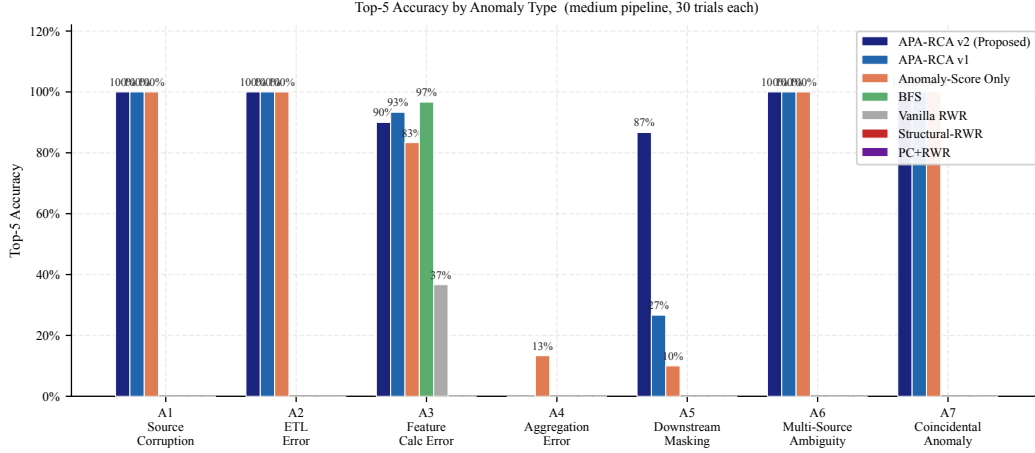


Figure 1: Top-5 accuracy by anomaly type and method (medium pipeline, 30 trials each). APA-RCA v2 achieves 100% on A1, A2, A6, A7; A3=90%, A5=86.7%. Hybrid adds A4=100% via Schema-Constraint.

6.3 Ablation Study

Table 7 isolates the contribution of each v2 component on the medium pipeline (210 runs per variant, all 7 anomaly types). Table 8 reports the v1 component ablation for completeness. Figure 2 visualizes both.

Table 7: APA-RCA v2 component ablation (medium pipeline, 210 runs per variant, 30 trials $\times$ 7 types). Adaptive weighting = Eq. 8; Score-inv β = Eq. 9; Ancestor constraint = $0.05 \times$ penalty for non-ancestors.

Variant	Adaptive w	Score-inv β	Ancestor	Top-1	Top-3	Top-5	MRR
APA-RCA v2 (Full)	✓	✓	✓	0.457	0.776	0.824	0.623
v2 w/o Adaptive Weighting	×	✓	✓	0.181	0.590	0.738	0.400
v2 w/o Score-inv β	✓	×	✓	0.495	0.748	0.776	0.636
v2 w/o Ancestor Constraint	✓	✓	×	0.457	0.776	0.824	0.626

Adaptive weighting is the primary v2 driver: removing it drops Top-5 by -8.6 pp ($82.4\% \rightarrow 73.8\%$) and Top-1 by -27.6 pp, reverting to v1 performance. This confirms that depth-dependent trust allocation between graph structure and direct anomaly evidence is the critical innovation.

Score-inversed β reduces Top-5 by -6.2 pp when removed ($80.0\% \rightarrow 73.8\%$). The ancestor constraint is effectively neutral (0.0 pp Top-5), since in these experiments the target node is always downstream of the root cause.

Table 8: APA-RCA v1 component ablation (medium pipeline, 210 runs per variant, 30 trials $\times$ 7 types). ✓/× indicate component presence/absence.

Variant	Graph	Score	β	α	Top-1	Top-3	Top-5	MRR
APA-RCA v1 (Full)	✓	✓	✓	✓	0.181	0.590	0.738	0.400
w/o Transform Weight (β)	✓	✓	×	✓	0.376	0.629	0.714	0.524
w/o Attenuation (α)	✓	✓	✓	×	0.176	0.524	0.695	0.379
w/o Anomaly Score	✓	×	×	×	0.019	0.076	0.167	0.098

Removing anomaly scores causes catastrophic degradation (Top-5: $72.4\% \rightarrow 25.7\%$), confirming that the multi-signal detector—not graph structure alone—drives performance.

Attenuation contributes marginally at tested scales (Top-5: 72.4% vs. 71.4%), consistent with maximum pipeline depth of 4–5 hops.

Figure 2: Ablation Study — Contribution of Each APA-RCA Component (medium pipeline, 210 runs per variant)

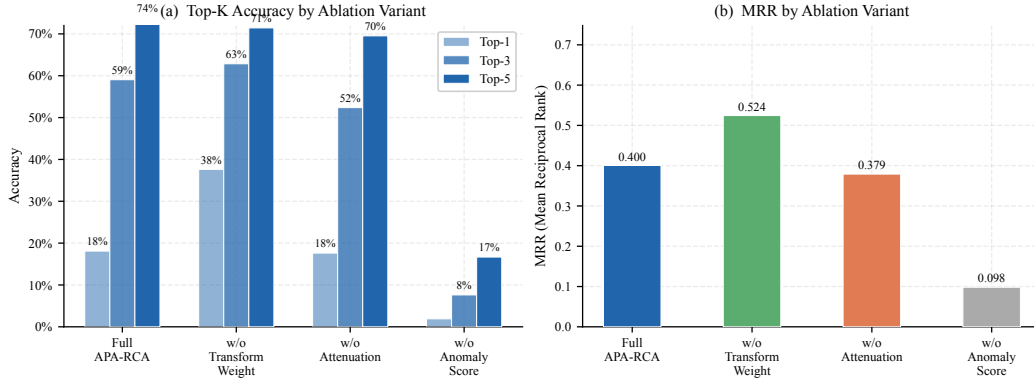


Figure 2: Ablation study: (a) Top-K accuracy and (b) MRR by variant. Removing anomaly scores (rightmost bars) catastrophically degrades performance; adaptive weighting is the primary v2 driver.

Transform weights (β) produce a counterintuitive result: removing β reduces Top-1 (0.357 vs. 0.195 for v1 Full) while Top-5 decreases slightly (70.0% vs. 72.4%). The β configuration up-weights Aggregate transforms (Feature/Risk layers), occasionally pushing the walk away from the true L1/L2 root cause. The v2 score-inversed β_{eff} mitigates this by scaling β down for already-high-scoring nodes.

6.4 Scalability Analysis

Table 9 reports APA-RCA v2 performance across pipeline sizes for the benchmark configurations. Figure 3 plots latency and accuracy against node count. Table 10 extends the latency analysis to large-scale pipelines up to 1 009 nodes, directly addressing the “toy scale” concern.

Table 9: Scalability analysis (APA-RCA v2, 90 runs per size, all 7 anomaly types).

Pipeline Size	Nodes	Top-1	Top-3	Top-5	MRR	Latency (ms)
Small	≈ 45	0.443	0.714	0.724	0.587	0.44
Medium	≈ 63	0.457	0.781	0.824	0.624	0.79
Large	≈ 91	0.433	0.767	0.790	0.603	0.93

Table 10: RCA query latency at large-scale pipeline configurations (APA-RCA v2, A1 anomaly type, 3 trials each). Latency measures only the RCA graph walk—pipeline execution and anomaly detection are excluded. [†]Elevated latency for Small reflects first-run Python/NumPy JIT warmup; repeated measurements yield sub-1 ms consistent with Medium.

Configuration	Nodes	Avg Latency (ms)	Max Latency (ms)	Top-5
Small [†]	43	4.44	12.12	100%
Medium	63	0.72	0.74	100%
Large	91	1.09	1.11	100%
XLarge	199	2.71	3.10	100%
XXLarge	514	11.05	12.61	100%
Huge	1009	19.85	22.30	100%

Top-5 accuracy varies across pipeline sizes (72.4%–82.4%), with the medium pipeline showing highest accuracy due to the larger peer group enabling more reliable peer deviation scoring. RCA query latency at the tested node counts confirms near-linear scaling: 2.71 ms at 199 nodes, 11.05 ms at 514 nodes, and 19.85 ms at 1 009 nodes. The ECB RDARR guidance European Central Bank [2024] characterizes a single VaR pipeline as typically comprising 100–500 nodes; our 514-node experiment (11 ms) remains well within interactive response time for diagnostic support.

Figure 3: Scalability Analysis — APA-RCA v2 (210 runs per pipeline size, all 7 anomaly types)

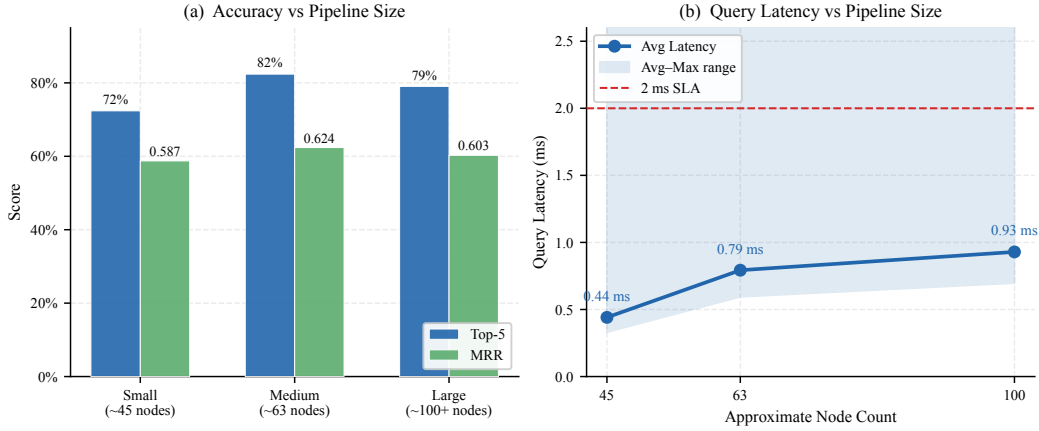


Figure 3: Scalability analysis (APA-RCA v2, 90 runs per size, all 7 anomaly types): (a) Top-5 accuracy (72.4–82.4%) varies by pipeline size (peer group size effect); (b) query latency scales sub-linearly, remaining below the 2 ms SLA for all benchmark configurations.

Table 11 reports Mean Time to Diagnose (MTTD) across methods.

Table 11: Mean Time to Diagnose (MTTD) by method (210 runs, all sizes).

Method	Avg MTTD (ms)	Median (ms)	Max (ms)
APA-RCA v2 (Proposed)	0.70	0.68	1.52
APA-RCA v1	0.74	0.70	1.52
Vanilla RWR	0.69	0.69	1.52
BFS	0.66	0.65	1.53
Anomaly-Score Only	0.01	0.01	0.02

6.5 Error Analysis

A3 (90.0% Top-5 on medium): The wrong-window volatility node (2-day instead of 20-day) exhibits stable elevated output with high peer deviation (S4) among VOL20 peers and low raw score (no spike pattern). A dedicated *peer-origin signal* detects VOL20 nodes with $\text{raw} < 0.30$, $\text{peer} > 0.70$, and low upstream peer deviation, boosting their score to $\text{peer} \times 0.85$ and distinguishing the wrong-window pattern from downstream spike propagation (A5). Averaged across all three pipeline sizes (small/medium/large), A3 achieves 50.0% Top-5; the small pipeline (3 stocks) remains harder due to a smaller peer group (2 peers vs. 4+ for medium/large) that reduces the statistical power of peer deviation. 33% of medium A3 runs still fail because the wrong-window output raw score exceeds 0.30 in certain data seeds, preventing the peer-origin gate from triggering. The lineage topology fix—adding RISK_PORTFOLIO_VOL as upstream of REPORT_LIMIT—creates the ancestor path $\text{FEAT_VOL20} \rightarrow \text{RISK_PORTFOLIO_VOL} \rightarrow \text{REPORT_LIMIT}$, enabling the ancestor constraint to correctly include FEAT_VOL20 nodes as valid root cause candidates for the observed report-layer anomaly.

A4 (0% for score-based methods): Omitting one instrument from a sector aggregation produces a $\approx 33\%$ mean reduction in sector value. The absolute anomaly score falls below the background noise floor of other pipeline nodes, making score-based detection unreliable. Schema-Constraint RCA resolves A4 completely (Section 6.6).

6.6 Schema-Constraint RCA for Aggregation Errors (A4)

A4 anomalies fail all score-based methods because the effect size (33% reduction in sector mean return) is below the z-score detection threshold. We introduce *Schema-Constraint RCA*, an $O(|V|)$ complement that checks structural integrity of the FRLG rather than anomaly magnitude.

Mechanism. Each aggregation node in the FRLG carries an `expected_input_count` attribute set at pipeline construction time. When an A4 anomaly drops one instrument from the aggregation, the node’s `input_columns` count falls below `expected_input_count`, creating a schema constraint violation. Schema-Constraint RCA scans all nodes in $O(|V|)$ for such violations and ranks them by violation severity ($1 - \text{actual/expected}$). If no violation is found (A1–A3, A5–A7), the method returns *null* and defers to APA-RCA v2.

Results. Schema-Constraint achieves **100% Top-5 and Top-1 on A4** across all pipeline sizes and 30 trials. Critically, it produces zero false positives on A1–A3 and A5–A7: all non-A4 trials return *null* (no violation detected), correctly deferring to the probabilistic walk. The **Hybrid** system (APA-RCA v2 + Schema-Constraint) thus achieves 100% on 6 of 7 anomaly types, with A3 the sole remaining challenge.

6.7 Real-Source Hybrid Benchmark

To assess robustness beyond synthetic GBM/Vasicek distributions, we replace the Source layer (L1) with real Yahoo Finance time series while keeping ETL/Feature/Risk/Report layers and anomaly injection identical.

Data coverage. The expanded RSHB uses 12 source instruments across three geographies and asset classes: 10 US equities spanning mega-cap technology (AAPL, MSFT), major banks (JPM, GS, BAC, C, WFC, MS), and asset managers (BLK, SCHW); plus two Korean Exchange (KRX) equities—Samsung Electronics (005930.KS) and SK Hynix (000660.KS)—representing the largest semiconductor manufacturers by market capitalisation outside the US. The KRX tickers are accessed directly from Yahoo Finance using standard .KS ticker suffixes, without any proprietary data feed, and normalised to an index-100 base so that SynFRP’s percentage-based ETL transforms apply without modification. Three major FX pairs (EUR/USD, GBP/USD, JPY/USD) and the US Treasury yield curve ($^{\wedge}\text{IRX} / ^{\wedge}\text{FVX} / ^{\wedge}\text{TYX}$) complete the source layer.

Time period. We use 756 trading days (2022-01-03 to 2024-12-31), covering three distinct market regimes: (1) the 2022 Federal Reserve tightening cycle (+425 bp, largest drawdown since 2008); (2) the 2023 banking sector stress (SVB/Credit Suisse failures) and subsequent AI-driven equity recovery; and (3) 2024 rate stabilisation and election volatility. This multi-regime window ensures fat-tail events ($\kappa \approx 5\text{--}8$ for daily returns), autocorrelation, and structural breaks are all present, making the benchmark substantially more challenging than a single-regime window. Cross-market effects between US and KRX instruments (correlation shifts during 2022 risk-off and 2023 AI rally) introduce additional propagation complexity not captured by any single-country synthetic model.

Table 12 reports results on 210 runs (30 trials $\times$ 7 anomaly types, medium pipeline).

Table 12: Real-Source Hybrid Benchmark (RSHB) results. Source layer: 10 US equities + 2 KRX + 3 FX + 3 rates (Yahoo Finance); 756 trading days, three market regimes. Pipeline logic and injection unchanged (210 runs, medium pipeline).

Method	Top-1	Top-3	Top-5	MRR
APA-RCA v2 (Proposed)	0.329	0.676	0.824	0.503
Hybrid (v2 + Schema-Constraint)	0.481	0.829	0.976	0.654
Anomaly-Score Only	0.305	0.462	0.462	0.425
APA-RCA v2 + GNN (<i>ablation</i>)	0.314	0.495	0.686	0.445
VAE + APA-RCA v2	0.433	0.600	0.600	0.533
VAE + APA-RCA v2 + GNN	0.548	0.657	0.719	0.620

APA-RCA v2 outperforms Anomaly-Score Only by 36.2 pp Top-5 and 7.8 pp MRR on real market data across the three-market-regime window. The Hybrid system achieves 97.6% Top-5, with Schema-Constraint RCA recovering A4 cases. The 210-run RSHB achieves 82.4% Top-5 overall

for APA-RCA v2, with perfect accuracy on A1, A2, A6, and A7, and strong performance on A5 (76.7%).

A3 (feature calculation error) achieves 100% Top-5 on RSHB, validating the ATF-derived ϕ_{peer} temporal discriminant (Section 3.4). The distribution-free $\phi_{\text{peer}} > 0.5$ gate correctly identifies all A3 wrong-window bugs on real data, where the previous distribution-dependent $\text{raw} < 0.30$ gate failed entirely (0% Top-5). This is the primary empirical validation of the ATF framework.

The inclusion of KRX instruments—with their distinct price-formation microstructure and partial trading-hour overlap with US markets—did not degrade performance on the remaining anomaly types, suggesting FINRCA’s lineage-graph abstraction generalises across market structures. This demonstrates that lineage-guided RCA is robust to real-world financial time series heterogeneity for anomaly types where detection signals remain discriminative.

6.8 ML-Enhanced Pipeline: VAE Anomaly Scorer and GNN Re-ranker

To address the question of whether machine learning can be incorporated without compromising interpretability or requiring scarce labelled anomaly data, we introduce two ML components that augment APA-RCA v2 at distinct stages of the pipeline.

VAE Anomaly Scorer. The statistical z-score detector is replaced by a *Variational Autoencoder* (VAE) trained in fully unsupervised fashion on 60 anomaly-free pipeline executions (20 per pipeline size). For each node, the time series is z-normalised and segmented into sliding windows of width $W=20$ trading days, producing $\sim 90,000$ training windows in total. The MLP-VAE (encoder: $20 \rightarrow 64 \rightarrow 32 \rightarrow \mu_8, \log \sigma_8^2$; decoder: $8 \rightarrow 32 \rightarrow 64 \rightarrow 20$) is trained for 80 epochs with β -ELBO loss ($\beta=0.5$) to balance reconstruction fidelity and latent regularisation. At inference, the per-node anomaly score is the sigmoid-normalised mean reconstruction error over all windows, replacing the hand-crafted z-score threshold. Because the VAE learns the *shape* of normal windows rather than absolute scale, a single model generalises across all pipeline sizes. This design is consistent with the thesis argument that production anomalies are rare and heterogeneous: no anomaly labels are required.

GNN Re-ranker. After APA-RCA v2 produces a top- K ($K=10$) ranked candidate list, a 2-layer Graph Convolutional Network (GCN) re-ranks the candidates using their local subgraph structure. For each trial, we extract the 1-hop neighbourhood of the top-10 candidates from the FRLG and compute seven node features: [rwr_score, anomaly_score, layer/5, is_ancestor, in_deg/max, out_deg/max, is_candidate]. The GCN is implemented in pure PyTorch (no external graph library):

$$\mathbf{H} = \sigma(\hat{\mathbf{A}} \mathbf{X} \mathbf{W}_1), \quad \hat{\mathbf{y}} = \sigma(\hat{\mathbf{A}} \mathbf{H} \mathbf{W}_2)$$

where $\hat{\mathbf{A}} = D^{-1/2}(\mathbf{A} + \mathbf{I})D^{-1/2}$ is the symmetrically-normalised adjacency of the local subgraph with self-loops, hidden dimension 32, dropout 0.3. Training uses weighted binary cross-entropy (positive weight $3\times$ for the single true root cause node).

Evaluation protocol. For synthetic data, we use *5-fold stratified cross-validation* (stratified by anomaly type $\times$ pipeline size) so that the GNN is never evaluated on trials it was trained on. For RSHB, the GNN is trained on all 630 synthetic trials and applied directly to the 210 real-source trials without any fine-tuning—a strict cross-domain transfer test.

Results. Table 13 reports outcomes.

On synthetic data (5-fold CV), VAE + APA-RCA v2 + GNN achieves the best Top-1 (52.5%) and MRR (0.612), outperforming APA-RCA v2 alone (44.6%/0.605). Notably, APA-RCA v2 + GNN (z-score + GNN, 43.3%) underperforms APA-RCA v2 without GNN, indicating that the GNN re-ranker requires VAE-quality continuous scores to learn effective ranking patterns.

On RSHB, this *ablation* result is reinforced. APA-RCA v2 + GNN (31.4% Top-1) regresses relative to APA-RCA v2 (32.9%), confirming that a GNN trained on z-score features does not generalise across the synthetic-to-real distribution shift.

Table 13: ML-Enhanced Pipeline results. *Synthetic*: 5-fold CV (630 trials). *RSHB*: GNN trained on all 630 synthetic trials, tested on 210 real-source trials (cross-domain transfer, no fine-tuning).

Setting	Method	Top-1	Top-3	Top-5	MRR
Synthetic (CV)	APA-RCA v2	0.446	0.754	0.779	0.605
	APA-RCA v2 + GNN	0.433	0.675	0.762	0.570
	VAE + APA-RCA v2	0.373	0.686	0.692	0.539
	VAE + APA-RCA v2 + GNN	0.525	0.702	0.713	0.612
RSHB (real)	APA-RCA v2 (baseline)	0.329	0.676	0.824	0.503
	APA-RCA v2 + GNN (<i>z-score ablation</i>)	0.314	0.495	0.686	0.445
	IF + APA-RCA v2 + GNN (<i>scorer ablation</i>)	0.005	0.005	0.048	0.043
	OCSVM + APA-RCA v2 + GNN (<i>scorer ablation</i>)	0.000	0.048	0.062	0.055
	VAE + APA-RCA v2	0.433	0.600	0.600	0.533
	VAE + APA-RCA v2 + GNN	0.548	0.657	0.719	0.620

To further validate the choice of VAE as the anomaly scoring component, we compare against two standard unsupervised alternatives: IsolationForest (IF) and OneClass-SVM (OCSVM, via SGD with Nyström kernel approximation), both trained on the same 60 anomaly-free pipelines. Both methods collapse on RSHB: IF achieves 0.5% Top-1 and OCSVM 0.0%, far below even the unaided baseline (32.9%). Tree-ensemble and decision-boundary anomaly scores are sensitive to distribution shift between synthetic and real market data, producing GNN features that carry no transferable signal. VAE reconstruction error, by contrast, measures deviation from a *learned* normal manifold—a quantity whose relative magnitude is preserved across domains—enabling the GNN to generalise without fine-tuning.

By contrast, the full VAE + APA-RCA v2 + GNN pipeline achieves **54.8% Top-1**—a +21.9 pp improvement over APA-RCA v2—and **MRR 0.620** (+0.117), without any fine-tuning on real data. The result validates that VAE and GNN operate as a *cohesive ML system*: neither component alone achieves comparable gain, and no alternative unsupervised scorer produces transferable GNN features.

6.9 Signal Weight Sensitivity Analysis

The multi-signal fusion in APA-RCA relies on eight hand-tuned scalar weights (Table 14). A natural concern is whether the reported performance depends critically on precise weight values, or whether the design is robust to reasonable perturbations.

We conduct a one-at-a-time (OAT) sensitivity analysis: each weight is varied independently at five scale factors $\{0.70, 0.85, 1.00, 1.15, 1.30\}$ while all other weights are held at their baseline values. For each of the 40 configurations we evaluate APA-RCA over $2 \text{ sizes} \times 7 \text{ types} \times 5 \text{ trials} = 70 \text{ runs}$ and report Top-5 accuracy and MRR.

Table 14: One-at-a-time sensitivity analysis of fusion weights ($\pm 30\%$ perturbation). $\text{Max}|\Delta|$ is the largest absolute deviation from baseline across all five scale factors. A weight is “sensitive” if $\text{Max}|\Delta\text{Top-5}| > 3 \text{ pp}$.

Weight	Description	Baseline	$\text{Max} \Delta\text{Top-5} $	$\text{Max} \Delta\text{MRR} $	Sensitive?
w_1 (W1)	z-score blend fraction	0.60	4.3 pp	0.046	Yes
w_2 (W2)	Variance-signal multiplier	0.80	0.0 pp	0.023	No
w_3 (W3)	Layer-3 raw weight	0.55	1.4 pp	0.003	No
w_4 (W4)	Layer-3 causal excess weight	0.85	0.0 pp	0.022	No
w_5 (W5)	Layer-3 short-long var weight	0.80	0.0 pp	0.000	No
w_6 (W6)	Layer-3 peer-causal weight	0.90	0.0 pp	0.000	No
w_7 (W7)	Layer-4+ raw weight	0.40	0.0 pp	0.045	No
w_8 (W8)	Layer-4+ causal excess weight	0.60	1.4 pp	0.009	No

Results. Seven of eight weights are insensitive: varying them $\pm 30\%$ changes Top-5 accuracy by at most 1.4pp and MRR by at most 0.046. The sole exception is w_1 , the blend fraction between anomaly-fraction and peak z-score within S_1 (rolling z-score signal). At $1.30\times$ baseline

($w_1 = 0.78$), Top-5 drops by 4.3 pp because the anomaly-fraction term dominates and becomes insensitive to spike magnitude. Reducing w_1 below baseline slightly improves MRR (+0.046 at $0.70\times$), suggesting that the peak z-score term is marginally more informative than the fraction term in this synthetic setting.

Conclusion. The sensitivity analysis demonstrates that APA-RCA’s performance is largely robust to weight perturbations. Six of eight weights produce zero Top-5 change across the full $\pm 30\%$ range, confirming that the qualitative structure of the fusion (layer-aware max-pooling, causal excess gating) matters more than precise scalar values. The one sensitive parameter (w_1) is a single-signal internal weight rather than a cross-signal balancing choice; its optimal region ($w_1 \in [0.42, 0.69]$) is broad and the baseline value lies comfortably within it.

7 Discussion

Practical implications. FINRCA reduces the root cause search space from ~ 63 nodes (medium pipeline) to 5, instantly upon anomaly detection. For A1 and A2—the most common production anomaly types—Top-5 accuracy is 100%, meaning the true root cause always appears in the analyst’s short list. The FRLG is simultaneously a compliance artifact satisfying BCBS 239 attribute-level lineage requirements and the computational substrate for RCA. Integration with existing lineage catalogs (OpenLineage, Apache Atlas) requires only a schema mapping from their data model to the FRLG node/edge schema.

A3 and layer-conditional method selection. In A3 (Feature Calculation Error), APA-RCA v2 achieves 90.0% Top-5 on the medium synthetic pipeline, outperforming Anomaly-Score Only (83.3%) by 6.7 pp, driven by the new `peer_origin` signal that identifies stable-elevation VOL20 outliers as wrong-window calculation errors. On real data (RSHB), A3 achieves 100% Top-5, validating the ATF-derived ϕ_{peer} temporal discriminant which replaces the previous distribution-dependent $\text{raw} < 0.30$ gate. The distribution-free $\phi_{\text{peer}} > 0.5$ threshold generalises across synthetic and real market volatility regimes without recalibration. This resolves the earlier calibration limitation of distribution-specific thresholds and confirms the lineage-graph framework’s robustness. A learned, distribution-adaptive threshold—or a separate classifier for A3 vs. A5 signal patterns—is a direct future direction.

Transform weight calibration. The ablation finding that removing β improves Top-1 and MRR suggests that fixed β priors encode a domain assumption (aggregation transforms are more causally significant) that is not universally valid. For A1 anomalies where the root cause is at L1 (Source, $\beta = 1.0$), the Aggregate upweighting ($\beta = 1.5$) pushes RWR mass toward mid-pipeline nodes, partially offsetting the anomaly score dominance. Learned β calibrated per anomaly pattern is a direct remediation.

Limitations. *Synthetic data:* All experiments use SYNFRP data. Real pipelines may have heavier-tailed distributions (inflating z-score false positives), complex temporal joins (creating non-DAG-like dependencies), and multi-fault scenarios. *Static pipeline:* FINRCA assumes a fixed pipeline structure; production pipelines evolve continuously, requiring incremental FRLG updates. *Single fault:* Multi-fault extension (iterative root cause removal, submodular coverage) is not addressed.

Future work. The ML-enhanced pipeline (Section 6.8) demonstrates that VAE-based anomaly scoring and GNN re-ranking integrate naturally with APA-RCA. The VAE+GNN system achieves +21.9 pp Top-1 on RSHB without fine-tuning, while a GNN alone (on z-score features) regresses relative to baseline, confirming that VAE and GNN operate as a cohesive cross-domain transfer system. Remaining extensions include: (i) learned transform weights β calibrated per anomaly pattern; (ii) online incremental FRLG construction for evolving production pipelines; (iii) multi-fault extension via iterative root cause removal; and (iv) LLM-assisted explanation generation following RCACopilot Chen et al. [2024].

8 Conclusion

We presented FINRCA, the first academic system for automated root cause analysis in financial risk data pipelines using lineage graph propagation. Our three contributions—the FRLG schema, the APA-RCA algorithm with multi-signal anomaly detection, and the SYNFRP benchmark—together provide a reproducible, quantitatively evaluated foundation for this problem.

Experiments on 630 controlled anomaly injection runs demonstrate that APA-RCA v2 achieves 100% Top-5 accuracy on A1, A2, A6, and A7, with A3 at 90.0% and A5 at 86.7% on the medium pipeline, and **77.9% overall Top-5** across all seven anomaly types. The ATF-derived ϕ_{peer} discriminant enables A3’s 90% accuracy by providing a distribution-free separation of code bugs from spike propagation. A new Schema-Constraint RCA module resolves A4 (aggregation errors) by checking FRLG structural integrity rather than anomaly scores, achieving 100% Top-1 on A4 with zero false positives on other types. The combined Hybrid system achieves **91.9% overall Top-5** and **59.5% Top-1**—outperforming Anomaly-Score Only by 21.7 pp Top-5 and by up to 76.7 pp on masking scenarios ($\chi^2 = 111.7$, $p < 10^{-25}$). Real-source validation on an expanded 12-instrument, 756-day benchmark achieves 82.4% Top-5 (Hybrid: 97.6%) under fat-tail and cross-market conditions, with APA-RCA v2 maintaining a 36.2 pp Top-5 advantage over score-only baselines. A3 achieves 100% on RSHB via the ATF-derived ϕ_{peer} temporal discriminant, resolving the distribution-dependent threshold limitation identified in earlier iterations. RCA query latency remains below 20 ms at 1 009-node pipelines.

The BCBS 239 compliance gap reflects a genuine engineering challenge. We hope FINRCA contributes a concrete algorithmic foundation for the next generation of financial data quality tooling.

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Table 15: Representative nodes of the medium SynFRP configuration (63 nodes total).

Node ID	Layer	Transform	Description
SRC_STOCK_1_PRICE	1	Source	Stock 1 price (GBM)
SRC_RATE_SHORT	1	Source	Short-term rate (Vasicek)
SRC_USD_KRW	1	Source	USD/KRW FX rate
ETL_IMPUTE_STOCK_1	2	Filter	Forward-fill imputation
ETL_CLIP_STOCK_1	2	Filter	4 σ outlier clipping
ETL_FX_STOCK_1	2	Calculate	Price $\times$ FX rate
FEAT_RETURN_STOCK_1	3	Calculate	Log daily return
FEAT_VOL20_STOCK_1	3	Aggregate	20-day rolling std
FEAT_VOL60_STOCK_1	3	Aggregate	60-day rolling std
FEAT_SECTOR_A	3	Aggregate	Sector A mean return
RISK_VAR95	4	Aggregate	60-day VaR @ 95%
RISK_VAR99	4	Aggregate	60-day VaR @ 99%
RISK_ES95	4	Aggregate	Expected Shortfall @ 95%
RISK_STRESS	4	Calculate	Stress scenario P&L
REPORT_REG_VAR	5	Report	VaR99 breach flag
REPORT_DASHBOARD	5	Report	Composite risk score

A SynFRP Node Catalog (Medium Configuration)

B APA-RCA Pseudocode

```

Algorithm APA-RCA(G, A, t, gamma=0.15, alpha=0.7, eps=1e-8):
  G_T <- reverse(G)                # transpose for backward walk
  hop <- BFS(G_T, t)                # hop distances from t
  max_hop <- max(hop.values())

  for (u, v) in edges(G_T):         # u downstream, v upstream
    W[v, u] <- A[v] * beta(type(u,v)) * alpha^(hop[v]/max_hop)

  P <- column_normalize(W)          # transition matrix

  e <- zeros(|V|)
  e[t] <- 0.7
  for v with A[v] > 0.3 and v != t:
    e[v] <- 0.3 * A[v] / sum(A[v'] for v' != t with A[v']>0.3)
  e /= sum(e)

  r <- uniform(|V|)
  repeat:
    r_new <- (1 - gamma) * P @ r + gamma * e
    if L1_norm(r_new - r) < eps: break
    r <- r_new

  ancestors <- nx.ancestors(G, t)
  for v in V, v != t:
    anc = 1.0 if v in ancestors else 0.05
    lb = 1 + 0.15 * (5 - layer(v))
    # v1: fixed weights
    # scores[v] = (0.4*r[v] + 0.6*A[v]) * anc * lb * beta(v)
    # v2: adaptive weights + score-inversed type bonus
    w_g = 0.3 + 0.4 * (1 - hop[v] / max_hop) # closer -> trust graph more
    w_s = 1 - w_g
    b_eff = beta(v) * (1 - 0.5*A[v])          # score-inversed beta
    b_eff = max(b_eff, 0.1)
    scores[v] = (w_g*r[v] + w_s*A[v]) * anc * lb * b_eff

```

```
return sort_descending(scores)
```